

10th October 2025

To whom it may concern,
Dear Sir/Ma,

RECOMMENDATION LETTER FOR MR. OGHENEFEGOR FAVOUR UGBINE

It is my pleasure to write a letter of recommendation for Mr. Oghenefegor Favour Ugbine for his future endeavours. I have had the opportunity to supervise Mr. Oghenefegor's master's thesis, while also serving as his lecturer in the Erasmus Mundus Joint master's in manufacturing 4.0 program, a prestigious scholarship sponsored by the European Commission for 25 students, of which Mr. Oghenefegor is a recipient. And I appreciate his intellectual rigour, technical prowess, and innovative spirit.

Mr. Oghenefegor's master's thesis, which he conducted as an intern at MISUTECH, titled "Identification by Inverse Method and Data Analysis of Mechanical Behaviour Laws for Numerical Simulation under Machining Conditions," represents a new contribution in the field of computational manufacturing. This topic was proposed by top industries where he tackled a fundamental and long-standing problem of the inability of numerical simulations to accurately predict machining conditions due to unreliable material data.

In his master's thesis, he set an original approach to this challenge. He developed and executed a highly computational model that coupled Python automation with Abaqus finite element simulations, utilizing a Levenberg-Marquardt optimizer. This wasn't just an existing tool, but it was a novel integration designed to inversely identify key material behaviour parameters, comprising a Johnson-Cook constitutive law and the thermomechanical behaviour called Taylor-Quinney, directly on machining data.

He is passionate about computational mechanics, manufacturing process modelling, and multiscale mechanics of materials, with a focus on integrating experimental and numerical approaches to create solutions for complex problems, utilizing his skill set. Beyond the technical qualities, Mr. Oghenefegor is a proactive, resilient, and collaborative researcher. He always engaged in insightful discussions. His ability to deeply understand complex mechanics, coupled with strong programming skills, is appreciated.

Through his master's thesis, he has also demonstrated a strong impact on the United Nations Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) by advancing sustainable manufacturing practices. His research directly supports the development of innovative computational tools that strengthen industrial processes, enhance resource efficiency, and promote resilience in modern manufacturing systems.

I recommend Mr. Oghenefegor Favour Ugbine without any reservation. I am convinced that he has the scientific acumen along with his technical and transversal skills to excel at all future endeavours.

Best regards,



Professor Frédéric Valiorgue